

Single-cell multiomic data analysis of brain

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Background

HIV in brain region can produce persistent molecular changes in the central nervous system, contributing HIV-associated neurocognitive disorders (**HAND**).

These heterogeneous effects may involve coordinated changes in **gene expression**, **chromatin accessibility** and **transcription factors(TFs)**.

- Up-regulation of immune/inflammation genes
- Down-regulation of transmission/signaling, and translation/metabolic genes

Additionally, clinical factors such as substance exposure, may further modify these regulatory programs and complicate the identification of disease-associated signals.

- Cannabis and Opioids proven to cause changes in GEX, and access to TFs and cCREs in the brain
 - Cannabis: Immune/Inflammation Genes
 - Opioids: KLF7, ADGRA3, TMEM164, etc.
 - 83.8% of PWH have used some form of illegal drugs

Goal

Using a newly available paired single-nucleus RNA-seq and ATAC-seq dataset, we aimed to identify HIV- and substance-associated transcriptional and regulatory changes.

We also developed a preprocessing pipeline that converts paired multiome data into cell-type-specific RNA and ATAC profiles for differential expression, differential accessibility analysis, and **GET** model input.

Data

SCORCH: Single Cell Opioid Responses in the Context of HIV
UM1DA051410–spudich

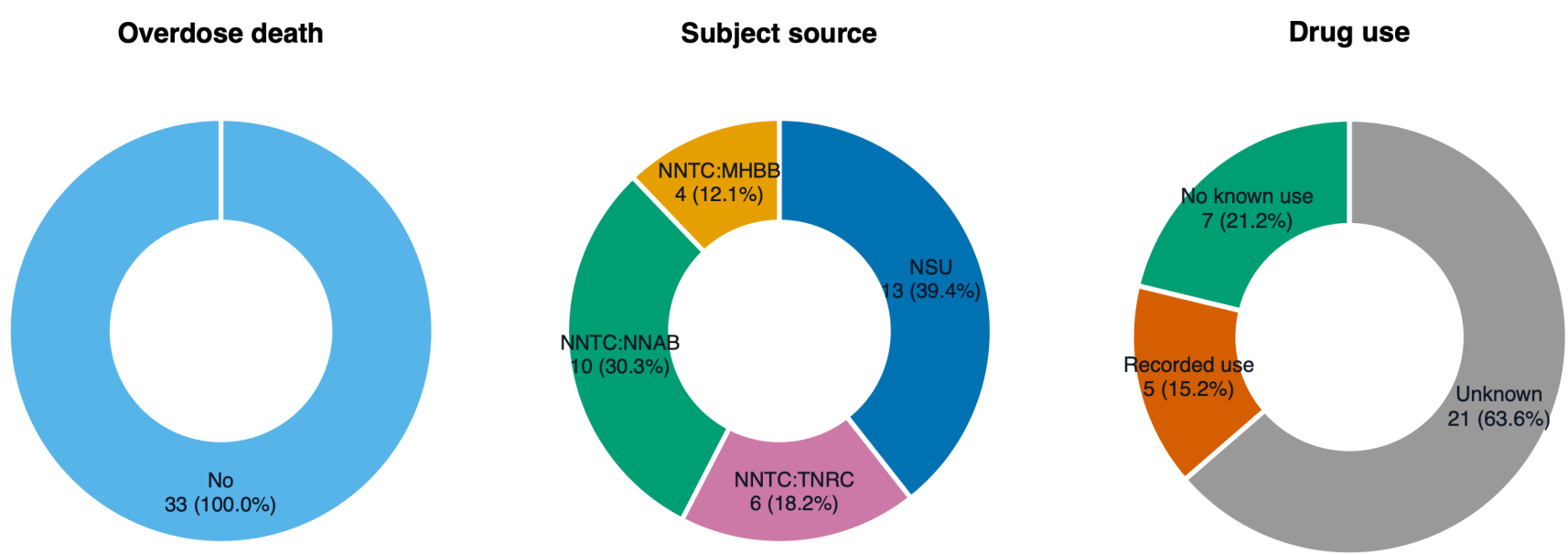
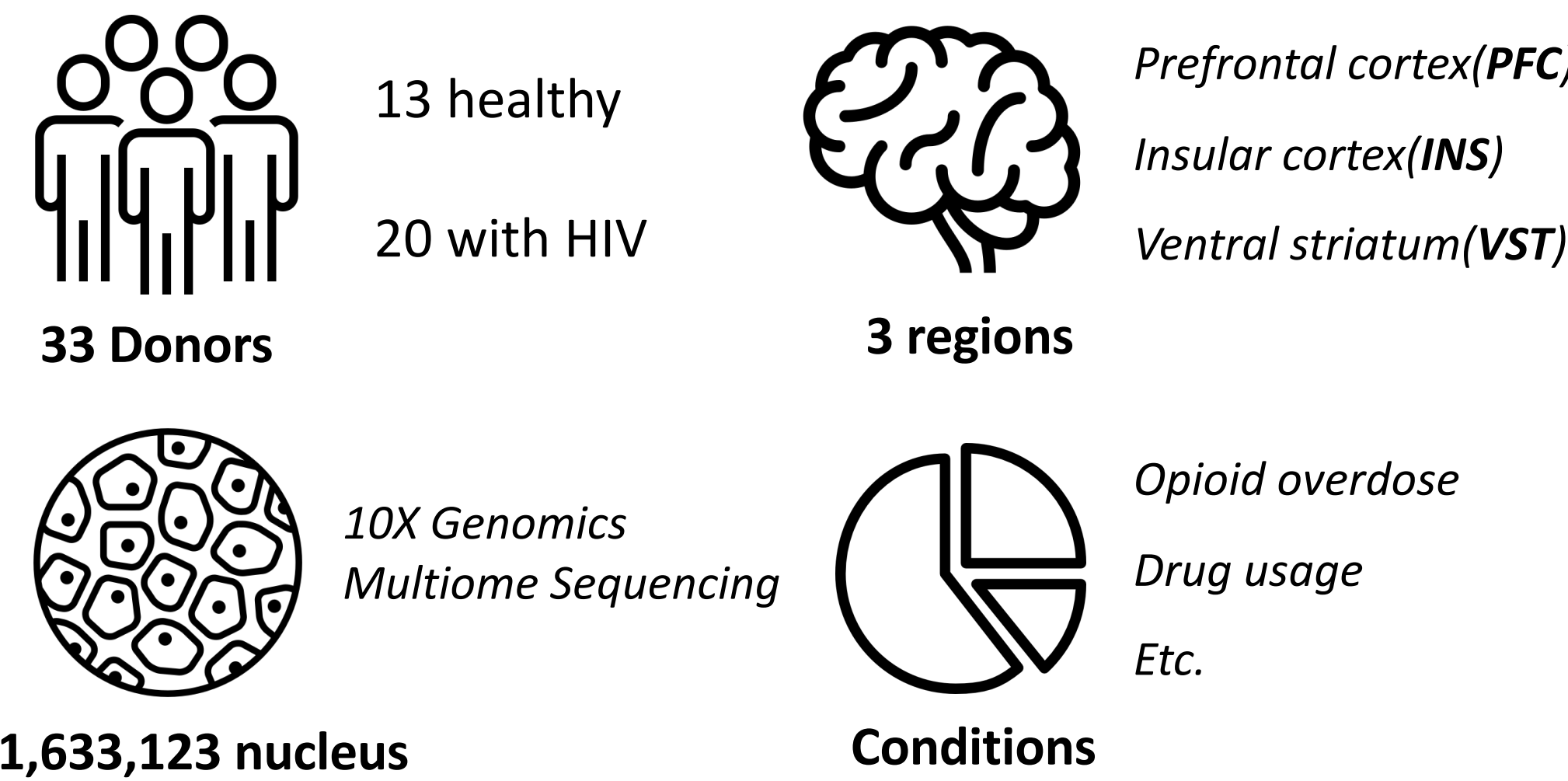


Figure 1. Cohort metadata

Paired data: scRNA-seq & scATAC-seq

Gene expression matrix: $\mathbf{X} \in \mathbb{N} * 0^{n \times G}$

ATAC peak count matrix: $\mathbf{A} \in \mathbb{N} * 0^{n \times P}$

where (n) is the number of paired cells, (G) is the number of genes, and (P) is the number of ATAC peaks.

The entries X_{ig} and A_{ip} denote the RNA count for gene (g) and the ATAC fragment count for peak (p) in cell (i), respectively.

Method

GET is a foundation model that learns how chromatin accessibility, transcription-factor motifs, and genomic context relate to gene expression.

We tested whether GET could recover the condition-associated signals identified by conventional RNA and ATAC analyses.

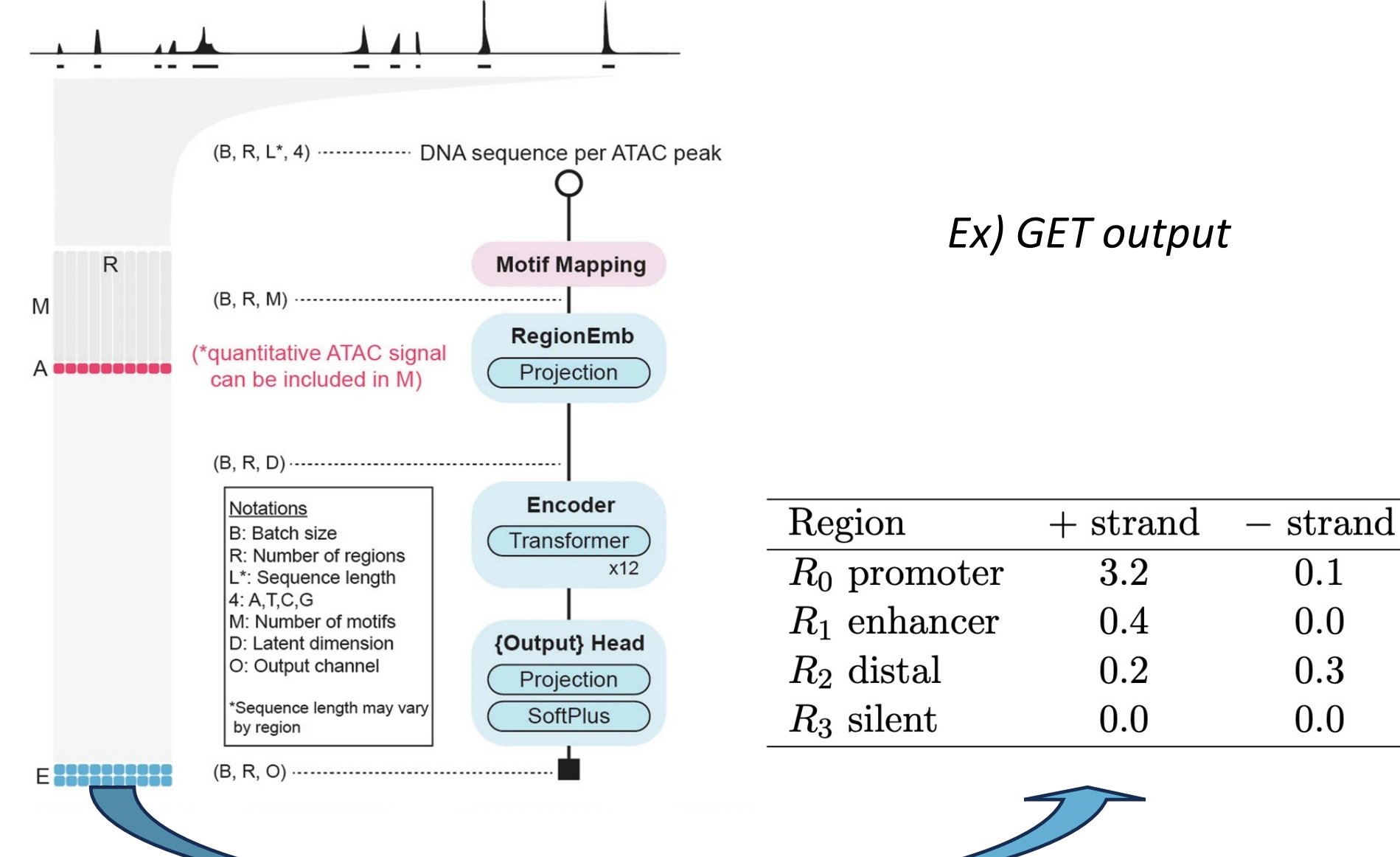


Figure 3. GET model architecture, output

When evaluated via external dataset's finetuned checkpoint, GET showed poor performance: $\text{Corr}(X_{pred}, X_{obs}) = 0.441$ underperforming our simple baseline predictors: TSS accessibility, gene activity.

Accordingly, its predicted gene expression has failed to differentiate HIV vs Healthy group, showing correlation values of near 1 between control, experimental group.

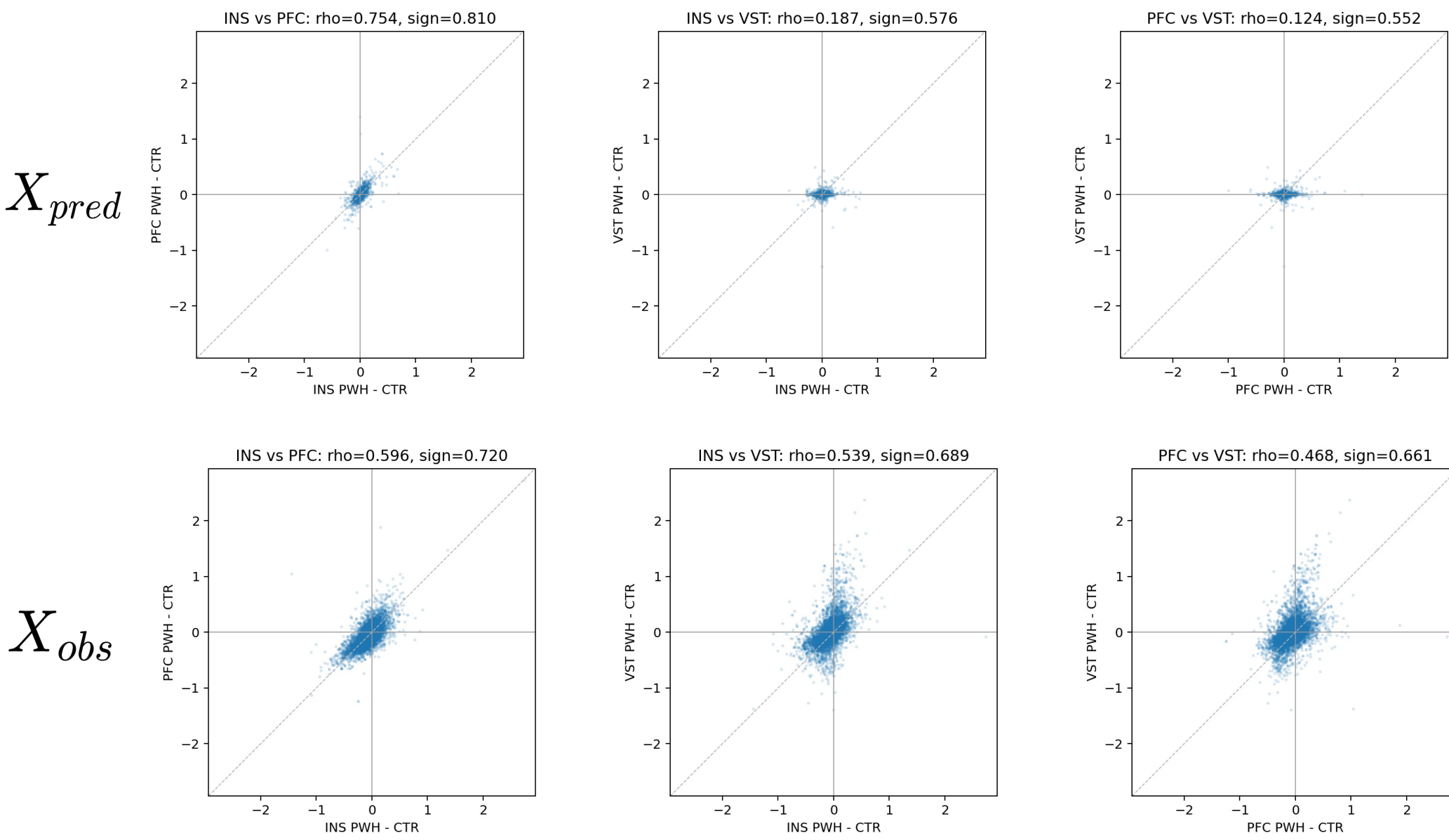


Figure 4. Pairwise scatter plot

Differential Analysis / Gene Ontology Analysis

- Identify each cell type
- Pseudo-bulk cells based on cell type
- Depending on comparison, group by condition
 - Virial Status: With or Without HIV
 - Brain region: Ventral Striatum (VST), Prefrontal Cortex (PFC), Insular Cortex (INS)
 - Drugs used: Alcohol/Opioid/Sedative, Alcohol/Opioid/Stimulant, Cannabis/Opioid, Heroin, Opioid
- Identify Differentially Expressed Genes using DESeq2
- Perform Gene Ontology Analysis using ToppGene's ToppFun

Results

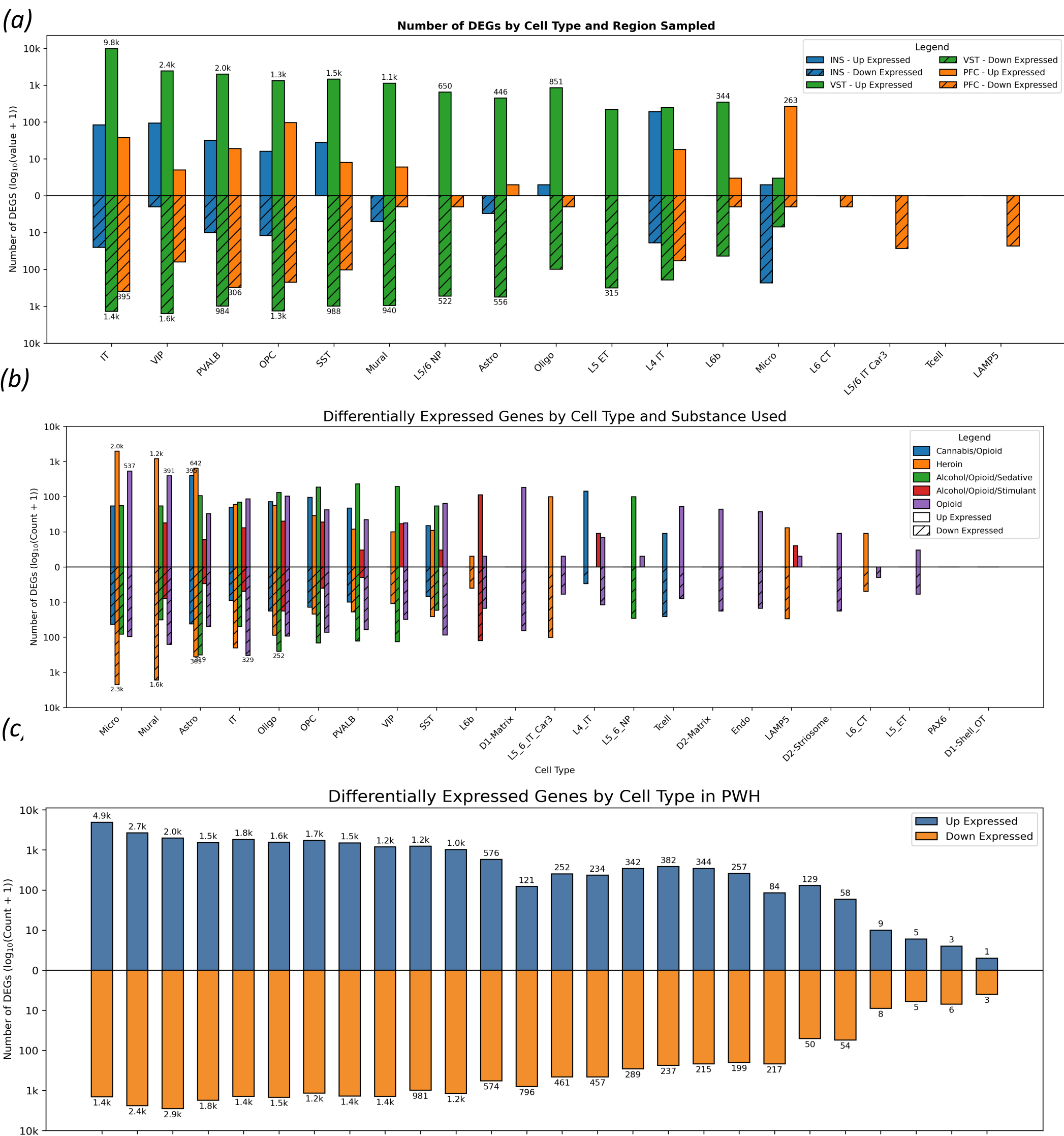


Figure 5. Differentially expressed genes by (a): region, (b): drug usage, (c) HIV condition

- VST significantly more up and down expressed DEG s than other regions when present
- HIV affected much more cell types than region and Substance Used
- Heroin and Opioid samples had much more DEGs than other substances

Common Biological Process/Molecular Functions of Differentially Expressed Gene sets

	Metabolic/ Translation	Transmembrane/ Signaling	Cell Structure	Immune	Cell Movement	Neuron Function/ Organization
PWH	9	9	2	2	0	0
Region Sequenced	0	13	0	0	1	2
Used Drugs	4	2	0	4	2	0

Figure 6. Table: GO analysis

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